SL2DT-05: Monitor & Alert Conversion

> **Series:** SL2DT | **Notebook:** 5 of 10 | **Created:** April 2026 | **Last Updated:** 04/21/2026

Overview

Goal of this step: rebuild Sumo Monitors in Dynatrace, choosing the right target for each (Anomaly Detection, Workflow with DQL threshold, or Metric Event). This is the notebook where the biggest fidelity-vs-better-approach decisions happen.

The anti-pattern to avoid: 1:1 static-threshold lift-and-shift. It produces noisy alerts, misses real anomalies, and under-uses Dynatrace's capabilities. The pattern to prefer: Anomaly Detection for metric-backed alerts, Workflows for complex multi-condition logic, and Metric Events for simple static thresholds that genuinely are static.

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Prerequisites

Requirement	Details
Audience	Core migration engineers + app-team reviewers
Inputs	`inventory/monitors.json`, translations from SL2DT-04
Dynatrace access	Platform Token with `settings:objects:write`, `automation:workflows:write`, `davis:anomaly-detectors:write`
Prior reading	SL2DT-04 for query translations; OPLOGS-09 for log-based alert patterns

1. What You'll Produce

Artifact	Purpose
`monitors/decision-matrix.csv`	Per-monitor: Dynatrace Intelligence vs Workflow vs Metric Event decision
`monitors/configs/`	Terraform + JSON for Dynatrace Intelligence detectors, Workflows, Metric Events
`notifications/action-map.md`	Sumo webhook/email → DT notification target map
`monitor-rebuild-report.md`	Progress per team

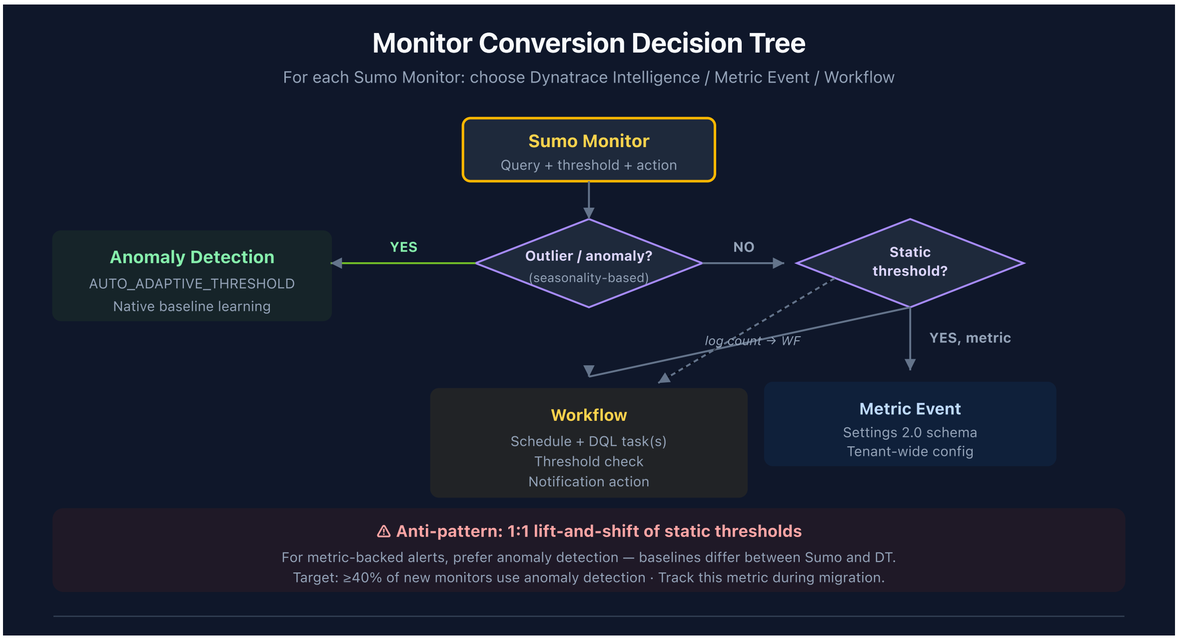
2. The Monitor Conversion Decision Framework

For each Sumo monitor, pick one of three targets:

Decision Rules

Sumo Monitor Type	Target	Why
Outlier / anomaly / seasonality-based	**anomaly detection**	Native baseline learning, no tuning

Sumo Monitor Type	Target	Why
Static threshold on metric (CPU > 90%)	**Metric Event**	Lightweight, tenant-wide config
Static threshold on log count	**Workflow with DQL**	Query runs on schedule, threshold in task condition
Missing data alert	**Workflow**	Use `isNull` / `count == 0` checks
Multi-condition (X AND Y but not Z)	**Workflow**	Multiple DQL tasks + branching
SLO-backed alert	**SLO burn-rate alert** (if Dynatrace SLO)	Native SLO product



3. Anomaly Detection — When & How

Dynatrace Intelligence learns the baseline of a metric and alerts when values deviate. Best for:

- Any Sumo outlier monitor (`outlier _count window=N threshold=K`)
- Static thresholds where the "normal" value drifts seasonally (weekly cycle, business hours)
- Static thresholds that fire too often in Sumo (false positives from workload swings)

Configuration

anomaly detection is configured via the Settings API (`builtin:anomaly-detection.metric-events`) or via the UI. For programmatic setup:

```
{
  "schemaId": "builtin:anomaly-detection.metric-events",
  "scope": "environment",
  "value": {
    "enabled": true,
    "summary": "API error rate anomaly",
    "queryDefinition": {
      "type": "METRIC_KEY",
      "metricKey": "custom.metric.api_error_rate",
      "aggregation": "AVG",
      "entityFilter": {
        "conditions": [
          {"type": "NAME", "operator": "CONTAINS", "value": "payments"}
        ]
      }
    },
    "modelProperties": {
      "type": "AUTO_ADAPTIVE_THRESHOLD",
      "alertCondition": "ABOVE",
      "alertingOnMissingData": false
    }
  }
}
```

Sumo outlier → anomaly detection example

Sumo:

```
_sourceCategory=prod/api | timeslice 1m | count | outlier _count window=10
threshold=3
```

Dynatrace Intelligence:
1. Create a custom metric from the DQL equivalent (via metric extraction in OpenPipeline, or via `makeTimeseries` into an extracted metric).
2. Configure anomaly detection on that metric with `AUTO_ADAPTIVE_THRESHOLD` mode.
3. Attach the Sumo monitor's notification channel to the detected problem.

Metric Extraction from Logs

If the Sumo monitor was log-count-based, extract a metric first:

```
{
  "schemaId": "builtin:logmonitoring.log-custom-metric",
  "scope": "environment",
  "value": {
    "key": "custom.metric.api_error_count",
    "enabled": true,
    "query": "dt.source_entity = \"prod/api\" AND loglevel = \"ERROR\"",
    "measure": {"type": "OCCURRENCE"},
    "dimensions": ["dt.entity.host"]
  }
}
```

Then apply anomaly detection to the extracted metric.

Validation

```
// Confirm metric extraction is producing data
timeseries c = avg(custom.metric.api_error_count), from:-1h, by:
{dt.entity.host}
| fieldsAdd c_max = arrayMax(c)
| filter c_max &gt; 0
```

4. Metric Events — Simple Static Thresholds

When the threshold truly is static and the underlying metric is well-defined:

- CPU > 90%
- Memory available < 5%
- Disk full
- Certificate expiry < 30 days

Use Metric Events via `builtin:anomaly-detection.metric-events`:

```
{
  "schemaId": "builtin:anomaly-detection.metric-events",
  "scope": "HOST-12345",
  "value": {
    "enabled": true,
    "summary": "Host CPU &gt; 90%",
    "queryDefinition": {
      "type": "METRIC_KEY",
      "metricKey": "dt.host.cpu.usage",
      "aggregation": "AVG"
    },
    "modelProperties": {
      "type": "STATIC_THRESHOLD",
      "threshold": 90.0,
      "alertCondition": "ABOVE",
      "violatingSamples": 3,
      "samples": 5
    }
  }
}
```

When NOT to use Metric Events

- Threshold depends on workload (use Dynatrace Intelligence)
- Condition involves multiple metrics (use Workflow)
- Alert requires custom notification payload (use Workflow)

5. Workflows — Complex or Multi-Condition Alerts

Workflows are the equivalent of Sumo's more complex monitors. Use when:

- Alert logic needs multiple DQL queries
- Conditional branches required
- Custom notification payload (ServiceNow incident with specific fields, etc.)
- Scheduled evaluation (every N minutes)

Workflow Structure

```
trigger:
  type: schedule
  cron: "*/5 * * * *"   # every 5 minutes
tasks:
  - name: check_error_rate
    type: dql_execution
    query: |
      fetch logs, from:-5m
      | filter dt.source_entity == "prod/api"
      | summarize total = count(), errors = countIf(contains(content,
"ERROR"))
      | fieldsAdd error_pct = 100.0 * toDouble(errors) / toDouble(total)
  - name: check_threshold
    condition: "{{ tasks.check_error_rate.records[0].error_pct }} &gt; 5"
    type: notification_servicenow
    payload:
      incident:
        short_description: "API error rate {{
tasks.check_error_rate.records[0].error_pct }}%"
        assignment_group: "Payments Platform"
        priority: 2
```

Scheduled Search → Workflow

A Sumo scheduled search (query + schedule + action) maps directly:

Sumo Scheduled Search Field	Workflow Equivalent
Query	`dql_execution` task
Schedule (run every N)	`trigger.schedule.cron`
Alert condition	`condition` on notification task
Webhook action	Notification task with target
Email action	`notification_email` task

DQL in Workflow Tasks

The translated DQL from SL2DT-04 goes directly here. Wrap long queries in `|` to preserve formatting:

```
tasks:
  - name: find_slow_transactions
    type: dql_execution
    query: |
      fetch logs, from:-5m
      | filter dt.source_entity == "prod/api"
      | parse content, "LD 'latency=' INT:latency"
      | filter latency &gt; 2000
      | summarize c = count(), by:{http.path}
      | sort c desc
      | limit 20
```

6. Rebuilding Notification Actions

Every Sumo monitor has one or more actions. Map each to a Dynatrace notification target.

Sumo Action	Dynatrace Target	Notes
Email	Workflow notification_email task	Same recipient list
Webhook → ServiceNow	Workflow notification_servicenow task	Rebuild incident payload
Webhook → Slack	Workflow notification_slack task	Rebuild message format
Webhook → PagerDuty	Workflow notification_pagerduty task	Rebuild event payload
Webhook → custom HTTP	Workflow HTTP task	
Mobile push	Workflow + Dynatrace Mobile app	

ServiceNow Integration — Specific Patterns

ServiceNow is the most common target and the highest-risk translation (wrong field mapping → misrouted incidents).

Sumo webhook payload:

```
{
  "incident": {
    "short_description": "{{Name}}: {{Description}}",
    "u_affected_service": "{{ClusterName}}",
    "assignment_group": "{{Team}}",
    "priority": "{{Priority}}"
  }
}
```

Dynatrace Workflow equivalent:

```
- name: create_servicenow_incident
  type: http
  method: POST
  url: https://{{instance}}.service-now.com/api/now/table/incident
  headers:
    Authorization: Basic {{secrets.servicenow_auth}}
  body:
    short_description: "{{ tasks.check_threshold.name }}: {{
tasks.check_error_rate.records[0].error_pct }}%"
    u_affected_service: "{{
tasks.check_error_rate.records[0].dt.entity.service }}"
    assignment_group: "Payments Platform"
    priority: 2
```

Verify with a test incident before flipping production traffic.

7. Handling Rare Alert Classes

Missing Data Alerts

Sumo:

```
_sourceCategory=prod/heartbeat | count
| where _count == 0
```

Dynatrace Workflow:

```
tasks:
  - name: check_heartbeat
    type: dql_execution
    query: |
      fetch logs, from:-10m
      | filter dt.source_entity == "prod/heartbeat"
      | summarize c = count()
  - name: alert_if_missing
    condition: "{{ tasks.check_heartbeat.records[0].c }} == 0"
    type: notification_email
    recipients: [oncall@example.com]
```

Change Detection Alerts

Sumo:

```
_sourceCategory=audit | logcompare timeshift=24h
```

Dynatrace: Change detection or two-fetch comparison workflow. See OPLOGS-08.

Multi-Metric Correlation

Sumo: two separate monitors with same target; Dynatrace: one Workflow with two DQL tasks + AND condition.

8. Tuning & Noise Reduction

Expect more alerts in the first week post-cutover — baselines aren't learned yet, and Workflow thresholds may be too aggressive.

First-week discipline

1. Monitor alert volume daily. Dashboard the total count of detected problems + Workflow triggers.
2. For every alert, triage: real issue, false positive, or threshold too tight?
3. Adjust thresholds daily until noise drops below Sumo baseline.

anomaly detection — let it learn

Dynatrace Intelligence needs ~2 weeks of data to build a stable baseline. During week 1–2, expect more sensitivity alerts. Keep them in a separate "tuning" severity class if possible; don't page on-call for them.

Checking alert volume

```
// Alert volume over time — compare Sumo baseline
fetch events, from:-7d
| filter event.kind == "DAVIS_PROBLEM"
| makeTimeseries problem_count = count(), interval:1d
```

Silence during load tests

Workflow-based monitors should respect maintenance windows. Settings 2.0 schema `builtin:maintenance.general` applies tenant-wide; configure before load tests, CI/CD deployments, or controlled outages.

Severity mapping

Sumo Priority	DT Severity	Notification
Critical	ERROR / Dynatrace Intelligence P1	PagerDuty + ServiceNow
High	Dynatrace Intelligence P2	ServiceNow
Warning	Dynatrace Intelligence P3	Email
Info	(no alert)	Logged only

9. Step Exit Criteria

G5 — Monitors Rebuilt

- [ ] Every Sumo monitor has a decision logged (Dynatrace Intelligence / Workflow / Metric Event / retire)

- [ ] ≥40% of new monitors use anomaly detection (not static thresholds) — track this metric
- [ ] Top-10 business-critical monitors validated end-to-end (fires on known-bad condition, doesn't fire on normal)
- [ ] Notification integrations (ServiceNow, Slack, PagerDuty) tested with live test incidents
- [ ] First-week alert volume ≤ Sumo baseline
- [ ] App teams trained on monitor rebuild (train-the-trainer — see SL2DT-01)

Next step: SL2DT-06 — Dashboard Conversion (rebuild dashboards using the translation output).
